FÍSICA REVISITADA · FR-TC-09
Research article

Memory and chance in the three-body problem: a planar test of the statistical theories

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We test, in the planar three-body problem with zero total angular momentum, whether the system forgets its initial conditions and with what measure it occupies the final states. In a map of 34,115 free falls with equal masses, the dependence of the outcome on the initial configuration disappears, within the resolution of the test, after five phases of democratic interaction. The energy distribution of the final pair agrees with the phase-volume theory, but the eccentricity distribution does not: the theory predicts 49% of pairs with eccentricity above 0.9, and the simulations give 35%. Since the deviation does not depend on the history, it is not memory but a failure of uniform occupation. We directly measure the absorptivity of Kol's flux theory in 210,000 time-reversed scatterings. The flux multiplied by it reproduces, with no fitted parameter, the distributions with equal masses and reduces the error in the ejection probabilities with unequal masses to about one percentage point, against 4.2 points for the volume theory. Unlike the three-dimensional case, the absorptivity is indispensable for predicting which body is ejected. The residual withstands a finer grid, a larger launch distance, and another chaos criterion, and it is smaller than the variation of the measured fractions themselves with the lifetime of the system, which indicates the limit of a time-independent statistical description.

Keywords: three-body problem; ergodicity; statistical theory; absorptivity; free fall; chaos.

ORCID: 0009-0008-2875-6607

Published: 3 November 2026. Version 2. Not peer reviewed.

Series: The three-body problem, part 9 of 9. https://fisica-revisitada.vercel.app/en/articles/memoria-e-acaso/

Translated from the original in Brazilian Portuguese.

I. INTRODUCTION

Since the trajectory of a non-hierarchical three-body system cannot be predicted by a formula, the remaining question is statistical: given the energy, the total angular momentum, and the masses, what is the probability of each outcome? This includes which body is ejected and with what energy and eccentricity the surviving pair is left. The classical approach assumes ergodicity, that is, that chaos scrambles the system to the point that it forgets its initial conditions and occupies with equal probability all final states compatible with the conservation laws [5, 8]. Stone and Leigh obtained in 2019 the closed form of this prediction, based on the phase-space volume [1]. Kol then proposed that the correct probability is not the volume but the flux of phase volume leaving the chaotic region, multiplied by an absorptivity: the probability that the same state, traversed backward in time, enters chaotic motion [2]. This factorization has been confirmed numerically in three dimensions, with equal masses and high angular momentum [3, 4].

This note performs two tests in the planar regime with zero total angular momentum, in which the volume theory agreed worst with the simulations [1] and in which, to our knowledge, Kol's factorization had not yet been tested with a measured absorptivity. In the plane with nonzero angular momentum, Dandekar and collaborators tested the flux theory without the absorptivity and found disagreement in the escape probabilities, which they attributed possibly to the failure of this simplification [10]. The first test asks where and when the system forgets where it started. The second compares the outcome distributions with three predictions: the phase volume, the pure flux, and the flux multiplied by the measured absorptivity. The method for measuring the absorptivity and the chaos criterion follow those of Manwadkar, Trani, and Kol [4]; what changes is the regime.

II. METHOD

A. Simulations and scrambles

We use the regularized integrator of the previous notes [7], with tolerance 10^{-12}, in planar motion with $G = 1$ and without softening of the potential, up to $1000\,t_{\mathrm{cr}}$. We call a scramble each passage from a state in which there is a bound pair with the third body more than two of its semimajor axes away to a state in which no pair satisfies this condition. The number of scrambles N_{scr} therefore counts the phases of democratic interaction, in which the three bodies interact without hierarchy, as in the criterion of democratic configurations in the literature [1, 4].

B. Memory map

In the free-fall problem, the three bodies start from rest, with zero total angular momentum. Fixing $A = (-0.5, 0)$ and $B = (0.5, 0)$, every initial configuration with equal masses corresponds, up to scale, rotation, and reflection, to a position of $C = (x, y)$ in the Agekyan-Anosova region, in which AB is the longest side and BC the shortest [6]. We cover this region with a grid of step 0.003, for a total of 34,115 initial conditions, and record for each one the ejected body, the energy and eccentricity of the final pair, the lifetime, and N_{scr}. To measure memory, we divide the map into cells of side 0.03 and test, for each range of N_{scr}, whether the distribution of which body leaves varies among the cells (chi-square test) and

whether the mean eccentricity varies among them (analysis of variance). A system that has forgotten where it started cannot show spatial dependence.

C. Theoretical predictions

Let $x = E_B/E_0$ be the energy of the final pair in units of the total energy and e its eccentricity. In the plane, the angular momentum l of the pair is a canonical variable, and the phase measure is uniform in l, which gives the factor $e/\sqrt{1-e^2}$. We redid the volume prediction of Stone and Leigh in a planar version: it weights each final state by the time the ejected body spends inside a strong-interaction region of radius $R = \alpha a_B(1+e)$, with $\alpha = 2$; the results barely change between $\alpha = 1$ and $\alpha = 5$. In Kol's theory, the weight of each state is the flux of phase volume through a distant surface, and the derivative of the energy with respect to the action of the ejected body cancels its period, so that, in the plane,

$$dP_s \propto k_s\, x^{-2}\, \frac{e}{\sqrt{1-e^2}}\, \mathscr{A}_s(x,e)\, dx\, de, \qquad k_s = \frac{(m_a m_b)^3}{m_a + m_b}, \quad (1)$$

where s indicates the ejected body, m_a and m_b are the masses of the pair, and $\mathscr{A}_s$ is the absorptivity. With $\mathscr{A}_s = 1$, we have the pure flux.

D. Measurement of the absorptivity

Reversing time, a final state becomes an incoming state: a pair with energy xE_0 and eccentricity e and a third body approaching on a hyperbolic orbit with angular momentum opposite to that of the pair, so that the total is zero. We launch the third body at a distance of 20 semimajor axes of the pair, with randomly drawn orbital phase and pair orientation, and measure the fraction of cases that enter chaotic motion, defined as $N_{\rm scr} \geq 2$ (we also tested $N_{\rm scr} \geq 3$). We use a grid of 7 values of x, between 1.05 and 10, and 12 values of e, between 0.05 and 0.995, with 400 runs per cell for equal masses and 300 per cell in each channel with unequal masses. In total, there were 33,600 runs for equal masses and 176,400 in seven channels with unequal masses. The absorptivity is interpolated between the grid points and extended above $x = 10$ by a power law.

E. Unequal masses

For the mass groups (1, 1, 3), (3, 3, 1), and (1, 2, 3), we ran 6,000 free falls per group, with initial positions drawn randomly in the unit disk as in note FR-TC-05 [9], and compared the ejection fraction of each body with the three predictions. The statistical uncertainty of the predictions with absorptivity was estimated by resampling the fractions measured in each cell.

III. RESULTS

A. Forgetting happens within a few scrambles

Within $1000\,t_{\rm cr}$, 97.8% of the free falls ended in ejection, and the relative energy error remained below 2×10^{-8}. Table I shows which body leaves as a function of the number of scrambles. Without scrambling, the outcome is almost determined by the initial geometry: body A, at the end of the longest side, leaves in 66% of the cases, and C, the free vertex, in only 1.4%. These cases, 12% of the map, form the regular regions visible in the Agekyan-Anosova region. The preference decreases with each scramble, and with 10 or more the distribution is close to one third for each body.

The cell test is more demanding, because it detects memory even when the global fractions are balanced. Considering all runs, the distribution of which body leaves varies strongly among the cells (chi-square per degree of freedom of 5.75), as does the mean eccentricity ($p \approx 10^{-188}$). With $N_{\rm scr} \geq 2$, the dependence is still significant (chi-square of 1.81 per degree of freedom; $p = 8 \times 10^{-11}$ for the eccentricity). With $N_{\rm scr} \geq 5$, it disappears: the chi-square is 1.06 per degree of freedom ($p = 0.13$) and the variation of the eccentricity among cells is not significant ($p = 0.27$). The same holds for $N_{\rm scr} \geq 10$. At the grid resolution, the map is almost entirely unpredictable: two neighboring points, separated by 0.003, have different outcomes in 62% of the cases, close to the 67% of a random draw.

B. Forgetting is not occupying everything equally

In the systems with $N_{\rm scr} \geq 5$, the energy distribution of the final pair agrees with the volume prediction, which in the plane gives a density proportional to $x^{-3.5}$ (Kolmogorov-Smirnov distance, KS, of 0.02 to 0.03). The eccentricity distribution, however, does not agree (Fig. 1). The volume prediction gives a mean eccentricity of 0.820 and 49% of pairs with $e > 0.9$; the simulations give 0.716 and 35% (KS = 0.18). This deviation does not decrease with more scrambles and does not depend on the position in the map, so it is not a memory effect: it is the hypothesis of uniform occupation that fails. The pure flux also fails, both in the eccentricity (mean of 0.786; KS = 0.12) and in the energy, which comes out too shallow (KS = 0.32).

C. Absorptivity

The measured absorptivity falls with the binding of the pair and with the eccentricity (Table II). A weakly bound, nearly circular pair is disrupted by the third body in 92% of the cases; a pair ten times more tightly bound, in 2% to 4%. Very eccentric pairs are absorbed less than circular ones with the same energy. Multiplying the flux by this absorptivity, with no fitted parameter, the prediction reproduces both measured distributions: in the eccentricity, a mean of 0.723 and KS of 0.018 with the criterion $N_{\rm scr} \geq 2$, or a mean of 0.718 and KS of 0.010 with $N_{\rm scr} \geq 3$, against 0.716 measured; in the energy, KS of 0.015

Table I. Fraction of ejections of each body in the free-fall map with equal masses, by number of scrambles. A and B are the ends of the longest side; C is the free vertex. Without memory, each fraction would be 1/3.

$N_{\rm scr}$	Runs	A leaves	B leaves	C leaves
0 to 1	4,032	0.657	0.328	0.014
2 to 4	7,178	0.351	0.402	0.247
5 to 9	7,579	0.351	0.351	0.298
10 or more	14,560	0.342	0.335	0.323

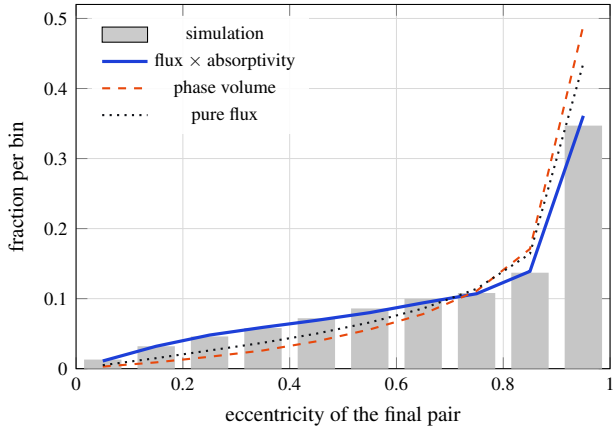


Fig. 1. Eccentricity distribution of the final pair in the free falls with equal masses and five or more scrambles (bars), compared with the planar version of the phase-volume prediction, with the pure flux, and with the flux multiplied by the measured absorptivity. None of the curves has a parameter fitted to the data.

to 0.022. The chaos criterion matters. If, instead of requiring democratic interaction, we count as chaotic every incoming state that takes more than one crossing time to resolve, the agreement worsens (KS of 0.063 in the eccentricity and of 0.146 in the energy). A delay without democratic interaction is, in general, a regular excursion, not chaos.

D. Which body is ejected with unequal masses

Table III compares the measured ejection fractions with the three predictions. The volume prediction exaggerates the advantage of the light body, as had already been observed in three dimensions [2, 3]: the heavy body leaves, for example, in 0.7% of the cases with (1, 1, 3), against 2.6% measured. The pure flux errs in the opposite direction, with 3.6%. In three dimensions, the pure flux gets the escape probability right to within 1% [3]. In the plane this does not happen, as Dandekar and collaborators also observed with nonzero angular momentum [10]; here, with zero angular momentum, the pure flux gives 5.1% for the heaviest body of the group (1, 2, 3), against 1.8% measured. The measured absorptivity corrects most of the deviations. The mean absolute error over the five fractions in the table is 4.2 percentage points for the volume, 2.5 for the pure flux, and 1.0 to 1.6 for the flux with absorptivity. The residual, of about one percentage point, is larger than the statistical uncertainty of the prediction, of 0.1 to 0.3 point, and is therefore systematic. In the group (1, 2, 3), the pure flux gets the fraction of the intermediate body right better than the prediction with absorptivity. The eccentricity distribution in the unequal groups is reproduced with KS of 0.03 to 0.07.

E. Origin of the residual

To locate the origin of the residual, we performed about 256,000 additional runs. First we tested three technical sources. The absorptivity grid was refined near the escape threshold, with 11 values of x between 1.01 and 20, because about 10% of the weight of the prediction lay below $x = 1.05$, a region that was previously extrapolated; the prediction changed by less than 0.5 point. The launch distance was doubled, from 20 to 40 semimajor axes, in the group (3, 3, 1); the predicted fraction of the heavy body went from 0.073 to 0.075. Finally, we adapted to the plane the chaos criterion of Manwadkar, Trani, and Kol [4], which counts nearly equilateral configurations before the first excursion; with equal masses, it worsened the agreement in the eccentricity (KS of 0.06 to 0.10, against 0.01 to 0.02 with the scramble criterion). None of these sources explains the residual.

The second part of the test examined the measured side. The ejection fraction depends on the history of the system. In the group (1, 2, 3), the probability that the light body leaves increases with the lifetime and that of the intermediate body decreases (rank correlation with $p = 0.005$ and $p = 0.014$); in the group (1, 1, 3), that of the heavy body increases ($p = 0.03$); in the group (3, 3, 1), the trend is not significant ($p = 0.19$). With equal masses, the mean eccentricity of the final pair rises from 0.689 in the systems that live less than $10\,t_{\rm cr}$ to 0.734 in those that live more than $100\,t_{\rm cr}$ ($p \approx 10^{-17}$). Therefore, the measured fraction depends on which runs are considered chaotic. We compared seven reasonable selections: at least two or at least five scrambles, short or long lifetime, with or without excursions, and at least two scrambles after the last excursion. Among them, the measured fraction varies on average by 3.4 percentage points, and all the flux-times-absorptivity predictions fall within this range (Table IV).

Table II. Absorptivity measured with equal masses (fraction of incoming states with two or more scrambles), for some values of the relative energy of the pair x and of the eccentricity e. Each value comes from 400 runs; the statistical error is at most 0.025.

x	$e = 0.05$	$e = 0.45$	$e = 0.85$	$e = 0.995$
1.05	0.92	0.61	0.41	0.44
1.5	0.48	0.33	0.21	0.16
3	0.30	0.11	0.07	0.07
10	0.04	0.03	0.02	0.02

Table III. Fraction of ejections per body in free falls with unequal masses and five or more scrambles (from 1,976 to 2,397 runs per group), with statistical error, compared with the planar version of the phase-volume prediction, with the pure flux, and with the flux multiplied by the measured absorptivity (range between the criteria of two and of three scrambles).

Masses	Ejected body	Measured	Volume	Pure flux	Flux × absorptivity
1, 1, 3	mass 3	0.026 ± 0.003	0.007	0.036	0.021 to 0.023
3, 3, 1	one of the masses 3	0.055 ± 0.005	0.022	0.100	0.072 to 0.074
1, 2, 3	mass 1	0.856 ± 0.007	0.936	0.821	0.829 to 0.841
1, 2, 3	mass 2	0.125 ± 0.007	0.053	0.128	0.139 to 0.151
1, 2, 3	mass 3	0.018 ± 0.003	0.012	0.051	0.020

IV. DISCUSSION

The results separate two questions that usually appear together. The first is whether the system forgets where it started. In the regime studied, the answer is yes, and quickly: five phases of democratic interaction suffice for neither the ejected body nor the shape of the final orbit to depend on the initial configuration, at the resolution of the test. The second is with what measure the system occupies the final states after forgetting, and here the answer is no: it is not the uniform measure of the phase volume. The deviation in the eccentricity is large, systematic, and independent of the history, which points to the premise of the volume theory and not to regions of regularity. In plain language, the randomness of three bodies is like a well-thrown but loaded die. The throw erases the initial position, and the bias is given by the absorptivity.

Kol's factorization passes the test in this regime. With equal masses, flux times absorptivity, with no free parameter, reproduces the energy and eccentricity distributions at the level of statistical precision, as already occurred in three dimensions with high angular momentum [4]. In the plane, the absorptivity does not cancel out in the escape probability, contrary to what occurs in three dimensions [3]: without it, the pure flux misses the fraction of the heavy body by a factor of 1.4 to 2.8. Dandekar and collaborators observed this disagreement in the plane with nonzero angular momentum and suggested that the cause was precisely the failure of the simplification without absorptivity [10]; the direct measurement made here, with zero angular momentum, confirms this explanation for most of the deviation.

The residual of about one percentage point does not come from the grid, the launch distance, or the chaos criterion. It is smaller than the variation of the measured fractions themselves with the duration of the interaction. Both theories, volume and flux, give a single distribution of outcomes, as if the system, after forgetting its start, always decayed in the same way. The simulations show that long-lived systems end slightly differently from short-lived ones, which is compatible with the observation that, with low angular momentum, more scrambles are needed for the system to become ergodic [11], and with the doubt, raised for the plane, about statistical equilibrium [10]. Kol's theory predicts this dependence through time-evolution equations that include the excursions [2]. As far as we could find, these equations have been compared with simulations only through the shape of the lifetime distribution [3, 10], and not through the change of the outcomes with the lifetime. In the regime studied, one percentage point seems to be the precision limit of a time-independent statistical description.

The main limitations are as follows. The simulations are planar and have zero total angular momentum; other regimes require new measurements. The memory map covers only free fall with equal masses, and a higher-resolution test could detect finer residual memory. The scramble criterion is an operational choice, and the quantitative conclusions depend on it, even though the criteria of two and of three scrambles give similar results. The absorptivity was measured with 200 to 400 runs per cell and extrapolated above $x = 20$. The correlations with the lifetime were measured in 6,000 runs per group, which suffices to detect trends of a few percentage points, but not to describe their shape.

V. CONCLUSIONS

In the planar three-body problem with zero angular momentum, the system forgets its initial configuration after about five phases of democratic interaction, but it does not come to occupy the final states equally: the phase-volume theory gets the energy of the final pair right and misses its eccentricity and the

Table IV. Ejection fraction predicted by the flux multiplied by the absorptivity, with the refined grid (range between the criteria of two and of three scrambles), compared with the fraction measured with five or more scrambles and with the range of values measured in the seven selections of runs described in the text.

Masses	Ejected body	Predicted	Measured	Measured range
1, 1, 3	mass 3	0.021 to 0.023	0.026	0.019 to 0.032
3, 3, 1	one of the masses 3	0.073	0.055	0.047 to 0.074
1, 2, 3	mass 1	0.833 to 0.844	0.856	0.797 to 0.860
1, 2, 3	mass 2	0.136 to 0.148	0.125	0.125 to 0.184
1, 2, 3	mass 3	0.019	0.018	0.015 to 0.021

ejection probabilities with unequal masses. Kol's flux theory, completed with the directly measured absorptivity, reproduces the distributions with equal masses with no fitted parameter and reduces the error in the ejection probabilities with unequal masses to about one percentage point. In this regime, unlike the three-dimensional case, the absorptivity is also indispensable for predicting which body is ejected. The remaining residual is not numerical: the outcomes depend slightly on the lifetime of the system, and a more precise description needs to take the time evolution into account.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

CITE AS

W. W. M. da Cunha, "Memory and chance in the three-body problem: a planar test of the statistical theories," Física Revisitada, Technical Note FR-TC-09 (2026), https://fisica-revisitada.vercel.app/en/articles/memoria-e-acaso/.

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